Chromatic Anchoring for Unsupervised Microstructural Analysis of Lightweight Concrete

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Presentation overview

01 Problem, prior approaches, and study scope

02 Experimental material and proposed workflow

03 Chromatic Anchoring and RGB K-means

04 Ablation, sensitivity, baselines, and efficiency

05 Exploratory extensions, limitations, and next validation steps

Microstructural porosity in lightweight concrete

Why it matters

- EPS and air-void fractions provide descriptors of material variability.
- The pore structure relates to density, transport behavior, and mechanical response.

Why current inspection is difficult

- Micro-CT requires suitable equipment and reconstruction procedures.
- Two-dimensional segmentation can change with lighting, surface preparation, and weak phase boundaries.

Study question: Can controlled exterior encoding reduce confusion between dark padding and internal voids in an unsupervised RGB workflow?

Related approaches and the study gap

Conventional characterization

- Optical microscopy and polished-section image analysis
- Thresholding, morphology, and watershed segmentation
- X-ray microtomography for three-dimensional characterization

Learning-based segmentation

- CNN and U-Net methods for cracks and air voids
- Performance depends on suitable annotations and acquisition conditions

Gap examined here

- Exterior padding can influence fitted RGB centroids when clustering includes those pixels.
- The paper examines red exterior encoding within a lightweight unsupervised pipeline.

The current experiments do not establish superiority over ROI-only fitting, other color spaces, or deep clustering methods.

Study scope and contributions

01 **Unsupervised workflow**

RGB K-means estimates EPS, matrix, and air-void area fractions without semantic labels for model fitting.

02 **Chromatic Anchoring**

A circular ROI and red exterior encoding reduce similarity between exterior padding and dark void pixels.

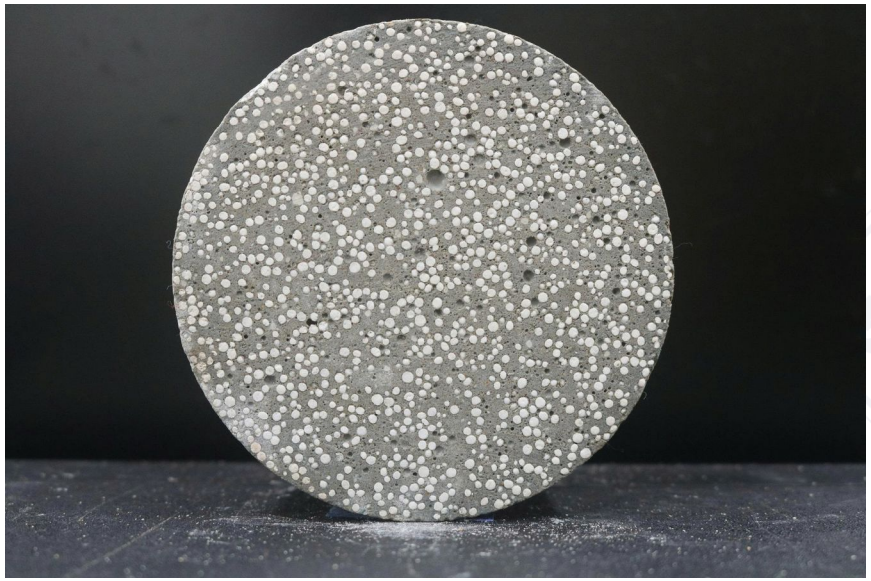
03 **CPU assessment**

The paper reports descriptive processing time and traced Python allocations on a laptop CPU.

04 **Exploratory analyses**

Synthetic perturbations, a porosity-strength association, and a serial-section illustration extend the feasibility study.

Experimental material and development collection



Representative polished cross-section

Property	Reported value
EPS particle size	1-3 mm
Nominal EPS volume fraction	60%
Fresh density	1554 kg/m³
28-day dry density	1395 kg/m³
28-day compressive strength	14.74 MPa
28-day water absorption	5.55%

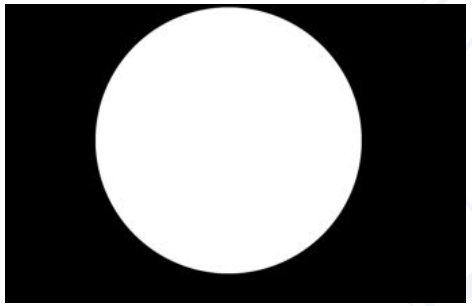
8 physical specimens, 8 cross-sectional images, 1920 × 1080 pixels

Proposed image-analysis workflow

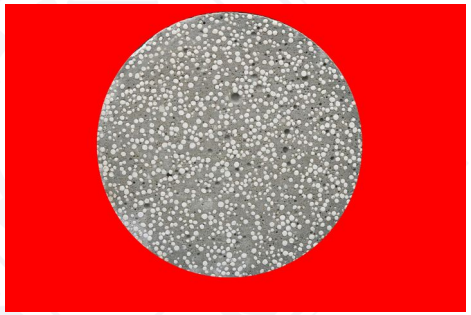
1 **Cross-sectional image**

Polish and image under controlled laboratory conditions

2 **Circular ROI mask**



3 **Red exterior encoding**



4 **RGB K-means and phase fractions**

Estimate EPS, matrix, and air-void area fractions

Chromatic Anchoring in RGB space

Dark exterior padding can resemble internal void pixels and influence K-means centroids when those pixels remain in the fit.



Black padding

RGB values may overlap with dark void appearance

Red anchor

a = (255, 0, 0)

RGB K-means and phase interpretation

Primary configuration

k = 3

Maximum 25 iterations



EPS particles

Relatively bright regions



Cementitious matrix

Intermediate appearance



Air voids

Relatively dark regions

- K-means uses 8-bit RGB pixel vectors.
- A consistent cluster-to-phase mapping supports comparisons across k = 3, 4, 5, and 6.
- Shadows, reflections, and mixed pixels can affect the appearance-based phase hypothesis.

Evaluation protocol and interpretation limits

Reference construction

- A primary researcher prepared masks in Roboflow.
- A senior domain expert reviewed ambiguous boundaries and adjudicated revisions.
- The study did not quantify independent inter-annotator agreement.

Reported metrics

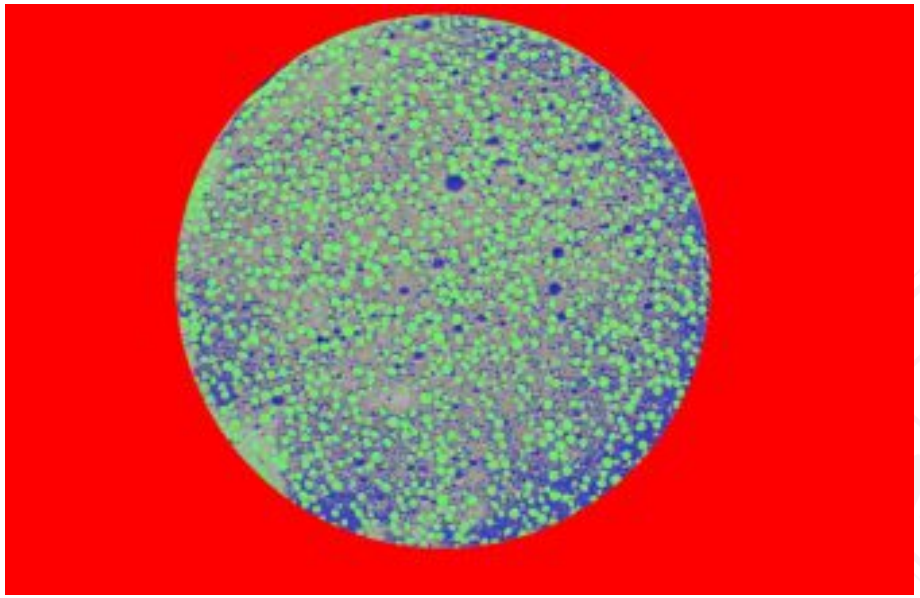
MAE — Mean absolute area-fraction error

RMSE — Root mean squared area-fraction error

DBI — Feature-space compactness and separation

Area-fraction agreement does not establish pixel-level overlap. Dice, IoU, and confusion metrics remain future evaluation targets.

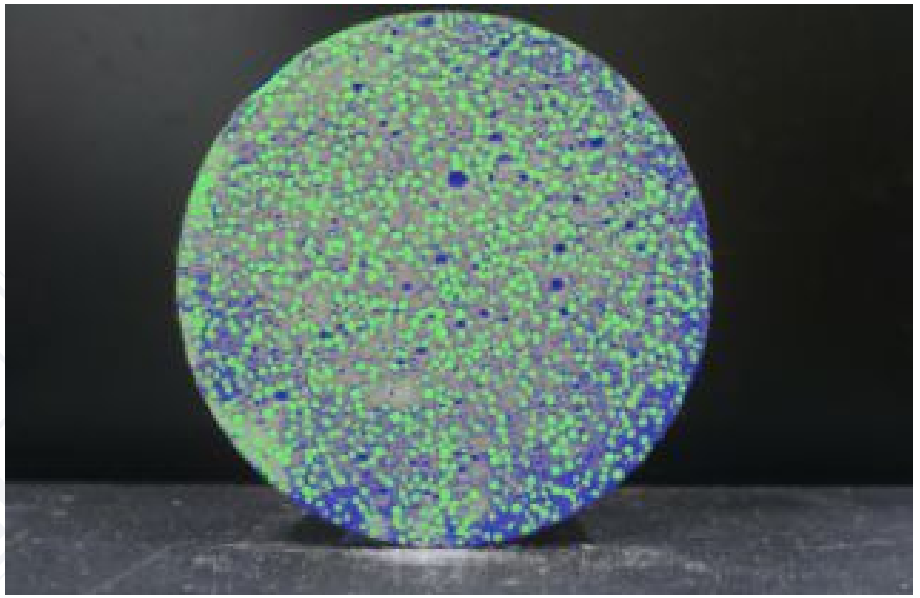
Exterior-padding ablation on one specimen



Chromatic Anchoring

1.39 pp

void-fraction error



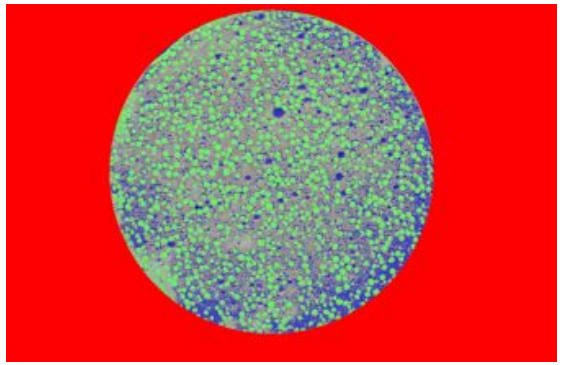
Conventional black padding

4.81 pp

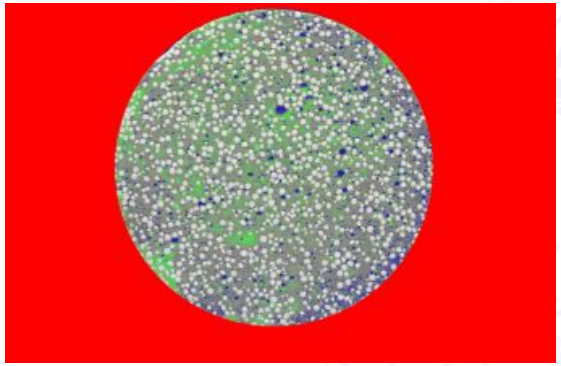
void-fraction error

Reference void fraction: 15.28% Red prediction: 16.67% Black prediction: 20.09%

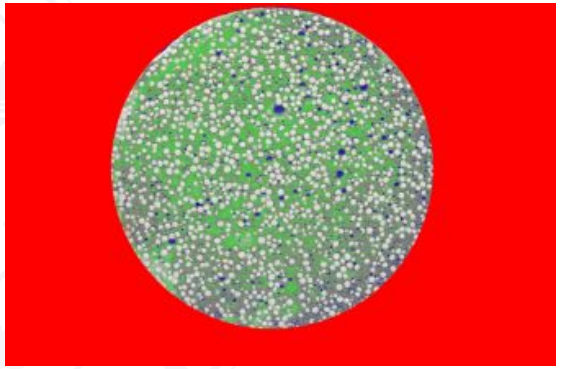
Qualitative sensitivity to the cluster count



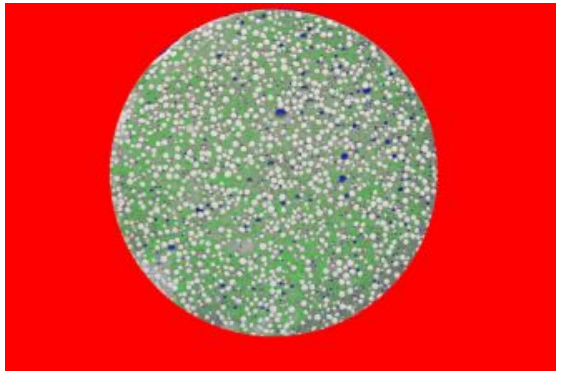
k = 3



k = 4



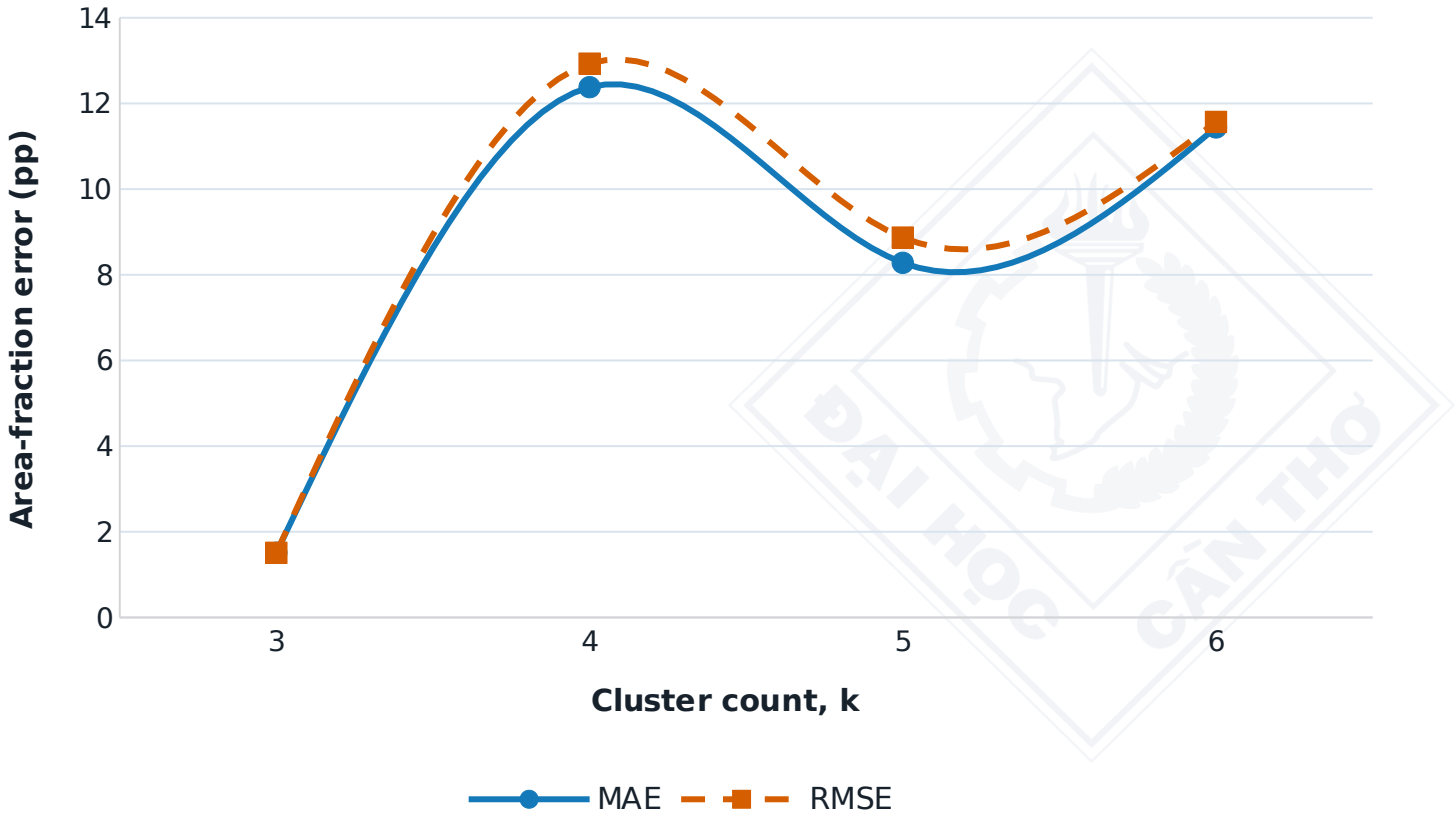
k = 5



k = 6

Green: EPS Gray: matrix Blue: voids Red: exterior

Quantitative sensitivity to the cluster count



k	MAE	RMSE
3	1.51	1.51
4	12.38	12.93
5	8.28	8.87
6	11.44	11.57

k = 3 yielded the lowest two-phase error among the tested settings

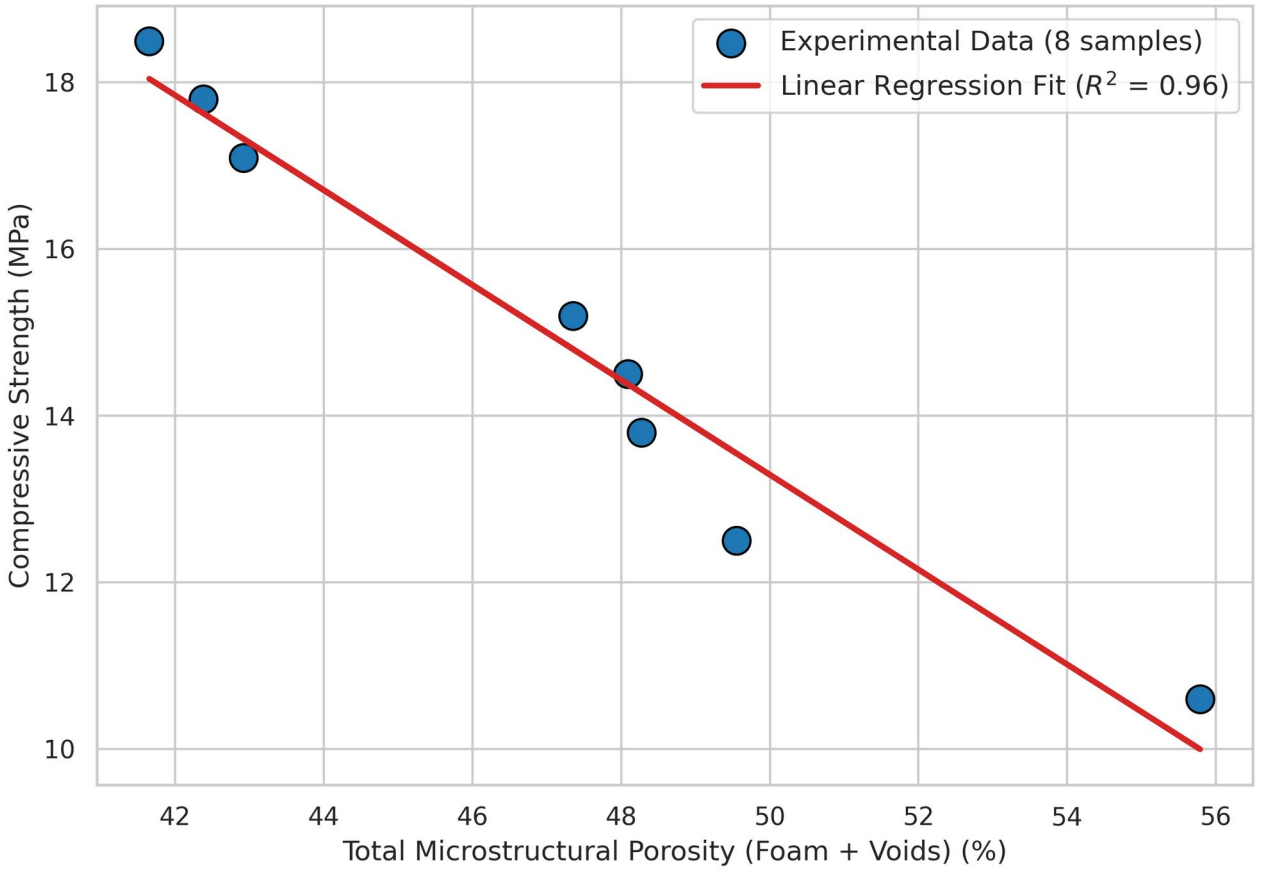
Exploratory comparison with baseline methods

Algorithm	DBI	MAE (pp)	Time (s/image)
Multi-level Otsu	0.5429 ± 0.0115	4.59	≈ 0.08
Gaussian Mixture	0.4605 ± 0.0234	-	≈ 14.02
Fuzzy C-Means	-	5.78	≈ 5.46
Deep Embedded Clustering	-	1.12	≈ 145.20*
K-means (k = 3)	0.5336 ± 0.0182	1.51	≈ 3.10

** DEC time excludes autoencoder pre-training.*

Missing measurements and non-harmonized timing prevent a definitive ranking.

Exploratory porosity-strength association



$R^2 = 0.9585$

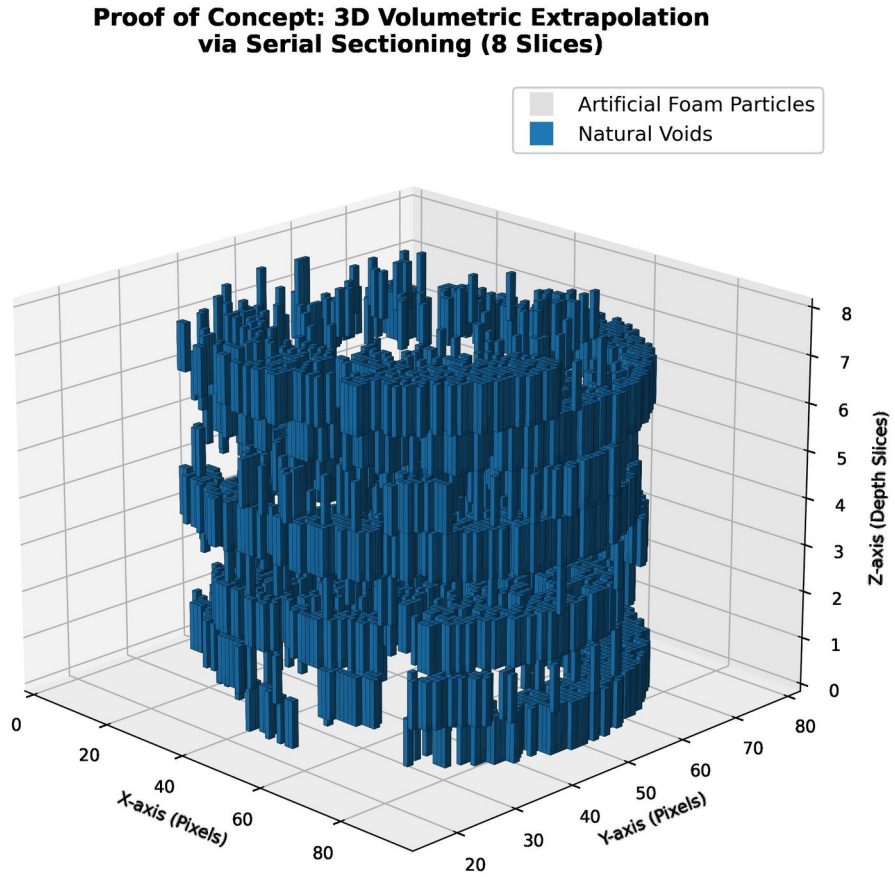
LOOCV

MAE 0.5757 MPa
RMSE 0.7336 MPa

n = 8 development specimens

Same observations used for fitting and internal resampling

Preliminary serial-section illustration



What the illustration shows

- Eight sequential two-dimensional sections
- Independent K-means segmentation for each section
- Label maps stacked along the depth axis

What remains unvalidated

- Inter-slice label consistency
- Absolute volumetric accuracy
- Comparison with an independent three-dimensional reference

Conclusion

Chromatic Anchoring reduced background–void confusion within a lightweight, unsupervised RGB K-means workflow.

What the study demonstrates

- Red exterior encoding reduced the representative void-fraction error from 4.81 to 1.39 pp.
- RGB K-means with k = 3 achieved a two-phase case-level MAE of 1.51 pp.
- CPU processing required about 3.10 s per 1920 × 1080 image, with approximately 17 MB of traced Python allocations.

Evidence to interpret cautiously

- Synthetic perturbations reached a maximum MAE of 2.15 pp on the same development images.
- The porosity–strength association was strong (R^2 = 0.9585), but it used only eight development specimens.
- No independent hold-out, inter-annotator agreement, or fully harmonized baseline benchmark was available.

The results support preliminary feasibility, not generalization or superiority across imaging conditions.

Future Work

01 Independent validation

Freeze the pipeline and evaluate a larger, held-out, multi-batch dataset spanning mixtures, laboratories, cameras, lighting, and surface preparation.

02 Reliable ground truth

Use a standardized multi-annotator protocol and report inter-annotator agreement together with Dice, IoU, and class-wise errors.

03 Controlled comparisons

Compare red encoding with ROI-only fitting, alternative anchors and color spaces, plus paired baselines under one timing and statistical protocol.

04 Deployment and 3D extension

Measure end-to-end memory and latency on target edge hardware, then validate serial-section estimates against micro-CT or another independent 3D reference.

Goal: establish when Chromatic Anchoring generalizes, where it fails, and whether it is reliable in field use.

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THANK YOU

Questions and discussion